YASKAWA

YRC1000micro OPTIONS
INSTRUCTIONS

FOR PMT FUNCTION

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

MOTOMAN-□□□ INSTRUCTIONS
YRC1000micro INSTRUCTIONS
YRC1000micro OPERATOR'S MANUAL
YRC1000micro MAINTENANCE MANUAL
YRC1000micro ALARM CODES (MAJOR ALARMS) (MINOR ALARMS)

The YRC1000micro alarm codes above consists of "MAJOR ALARMS" and "MINOR ALARMS".

Please have the following information available when contacting Yaskawa Customer Support:

- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
  Robot Serial Number *(Located on robot data plate)*
  Robot Sales Order Number *(Located on controller data plate)*

Part Number: 181284-1CD
Revision: 0

Yaskawa Customer Support

- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
HW1484458

DANGER

- This manual explains the PMT function (Position Modification Function for Tool Deformation) of the YRC1000micro system. Read this manual carefully and be sure to understand its contents before handling the YRC1000micro. Any matter not described in this manual must be regarded as "prohibited" or "improper".
- General information related to safety are described in "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS. To ensure correct and safe operation, carefully read "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS.

CAUTION

- In some drawings in this manual, protective covers or shields are removed to show details. Make sure that all the covers or shields are installed in place before operating this product.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids the product warranty.

NOTICE

- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the YRC1000micro.

In this manual, the Notes for Safe Operation are classified as “DANGER”, “WARNING”, “CAUTION”, or “NOTICE”.

DANGER

Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. Safety Signs identified by the signal word DANGER should be used sparingly and only for those situations presenting the most serious hazards.

WARNING

Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury. Hazards identified by the signal word WARNING present a lesser degree of risk of injury or death than those identified by the signal word DANGER.

CAUTION

Indicates a hazardous situation, which if not avoided, could result in minor or moderate injury. It may also be used without the safety alert symbol as an alternative to “NOTICE”.

NOTICE

NOTICE is the preferred signal word to address practices not related to personal injury. The safety alert symbol should not be used with this signal word. As an alternative to “NOTICE”, the word “CAUTION” without the safety alert symbol may be used to indicate a message not related to personal injury.

Even items described as “CAUTION” may result in a serious accident in some situations.
At any rate, be sure to follow these important items.

NOTE

To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” and “CAUTION".

DANGER

- Before operating the manipulator, make sure the servo power is turned OFF by performing the following operations. When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.
  - Press the emergency stop button on the programming pendant or on the external control device, etc.
  - Disconnect the safety plug of the safety fence. (when in the play mode or in the remote mode)

If operation of the manipulator cannot be stopped in an emergency, personal injury and/or equipment damage may result.

Fig. : Emergency Stop Button

- Before releasing the emergency stop, make sure to remove the obstacle or error caused the emergency stop, if any, and then turn the servo power ON.

Failure to observe this instruction may cause unintended movement of the manipulator, which may result in personal injury.

Fig. : Release of Emergency Stop

- Observe the following precautions when performing a teaching operation within the manipulator's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the manipulator’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the manipulator, which may result in personal injury.

- Confirm that no person is present in the manipulator's operating range and that the operator is in a safe location before:
  - Turning ON the YRC1000micro power
  - Moving the manipulator by using the programming pendant
  - Running the system in the check mode
  - Performing automatic operations

Personal injury may result if a person enters the manipulator's operating range during operation. Immediately press an emergency stop button whenever there is a problem. The emergency stop button is located on the right of the programming pendant.

- Read and understand the Explanation of the Warning Labels before operating the manipulator.

DANGER

- In the case of not using the programming pendant, be sure to supply the emergency stop button on the equipment. Then before operating the manipulator, check to be sure that the servo power is turned OFF by pressing the emergency stop button.
  Connect the external emergency stop button to the 4-14 pin and 5-15 pin of the Safety connector (Safety).
- Upon shipment of the YRC1000micro, this signal is connected by a jumper cable in the dummy connector. To use the signal, make sure to supply a new connector, and then input it.

If the signal is input with the jumper cable connected, it does not function, which may result in personal injury or equipment damage.

WARNING

- Perform the following inspection procedures prior to conducting manipulator teaching. If there is any problem, immediately take necessary steps to solve it, such as maintenance and repair.
  - Check for a problem in manipulator movement.
  - Check for damage to insulation and sheathing of external wires.
- Return the programming pendant to a safe place after use.

If the programming pendant is left unattended on the manipulator, on a fixture, or on the floor, etc., the Enable Switch may be activated due to surface irregularities of where it is left, and the servo power may be turned ON. In addition, in case the operation of the manipulator starts, the manipulator or the tool may hit the programming pendant left unattended, which may result in personal injury and/or equipment damage.

Definition of Terms Used Often in This Manual

The MOTOMAN is the YASKAWA industrial robot product.

The MOTOMAN usually consists of the manipulator, the YRC1000micro controller, manipulator cables, the YRC1000micro programming pendant (optional), and the YRC1000micro programming pendant dummy connector (optional).

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
YRC1000micro controller	YRC1000micro
YRC1000micro programming pendant	Programming pendant (optional)
Cable between the manipulator and the controller	Manipulator cable
YRC1000micro programming pendant dummy connector	Programming pendant dummy connector (optional)

Descriptions of the programming pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. ex. [ENTER]
	Axis Keys /Number Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, ex. [SHIFT]+[COORD]
	Mode Key	Three kinds of modes that can be selected by the mode key are denoted as follows: REMOTE, PLAY, or TEACH
	Button	Three buttons on the upper side of the programming pendant are denoted as follows: HOLD button START button EMERGENCY STOP button
	Displays	The menu displayed in the programming pendant is denoted with { }. e.g. {JOB}
PC Keyboard		The name of the key is denoted. e.g. Ctrl key on the keyboard

Description of the Operation Procedure

In the explanation of the operation procedure, the expression "Select • • • " means that the cursor is moved to the object item and [SELECT] is pressed, or that the item is directly selected by touching the screen.

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or brand names for each company or corporation. The indications of (R) and TM are omitted.

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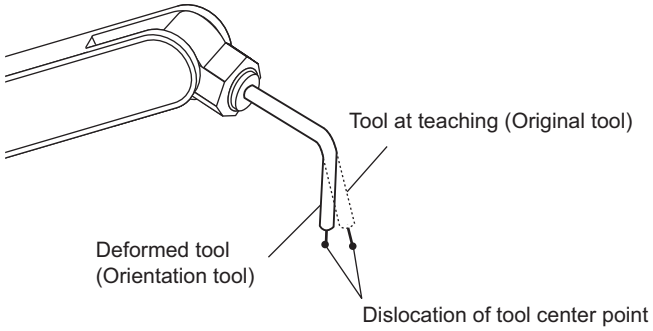
1 PMT Function

The PMT function corrects the position data when the tool is deformed during operation
(PMT: Position Modification for Tool Deformation). When a tool inadvertently collides with peripheral jigs or walls during operation and is deformed, the tool center point will be dislocated and the taught position of the job programmed would be shifted; it requires a great deal of time and effort for correction. In such a case, the PMT function can be used to correct the position data easily and accurately.

By specifying the desired job, the position data of the taught tool dimensions before deformation are converted automatically to the position data of the tool dimensions after deformation.

In the PMT function, the tool before deformation is called “the original tool” and the tool after deformation is called “the orientation tool”.



NOTE

- For a job in which teaching is performed by using several types of tools, only the position data obtained by the specified original tool is converted.
- The position variables are not converted in the PMT function.

2 Data Conversion by PMT Function

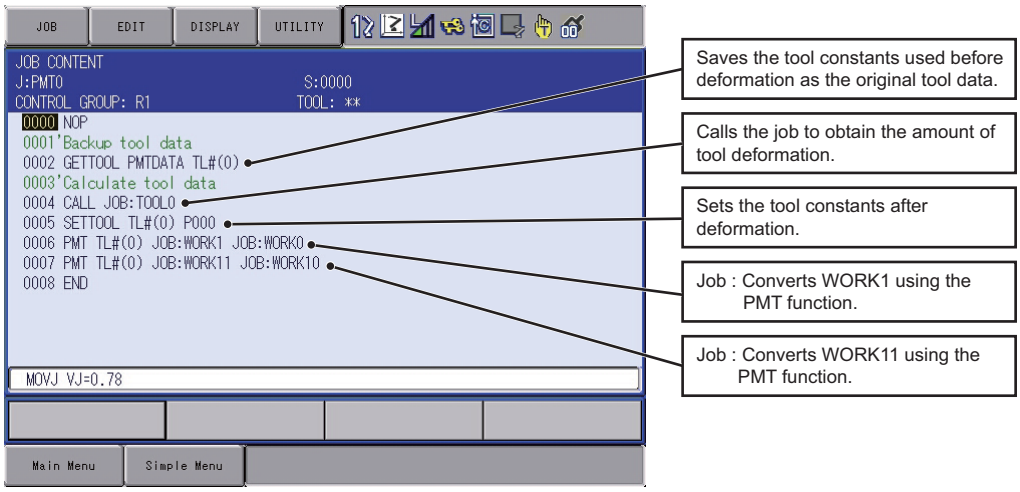
2.1 Outline

> **NOTE**
> When the PMT function is used, the tool dimensions and the job taught position data are rewritten. In order to secure the data, prepare the backup files beforehand.

The data can be converted by the PMT function using instructions or the programming pendant.

2.1.1 PMT data conversion by instruction

The following window is a job example (PMT0) to convert the data by instructions.
Carry out Job PMT0 to convert the data using the PMT function.



Refer to *chapter 2.2 "PMT Data Conversion by Instruction"* for the programming of each instruction, GETTOOL, SETTOOL, and PMT. Before converting the data, prepare a job, like TOOL0 in the preceding example, to obtain the amount of tool deformation by using a touch sensor or other device.

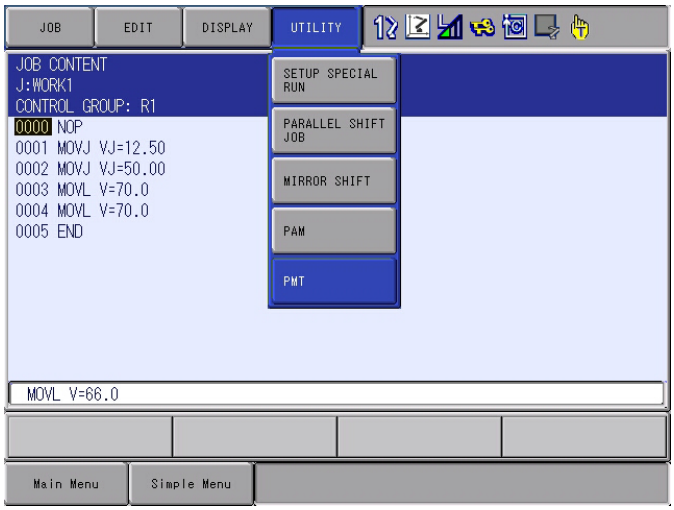
> **NOTE**
> - Before converting the data by a PMT instruction, be sure to save the data of the original tool by using a GETTOOL instruction.
> - For a system with the independent control function enabled, do not use PSTART, an independent control instruction, to start a job that includes a PMT instruction.

2.1.2 PMT data conversion by programming pendant

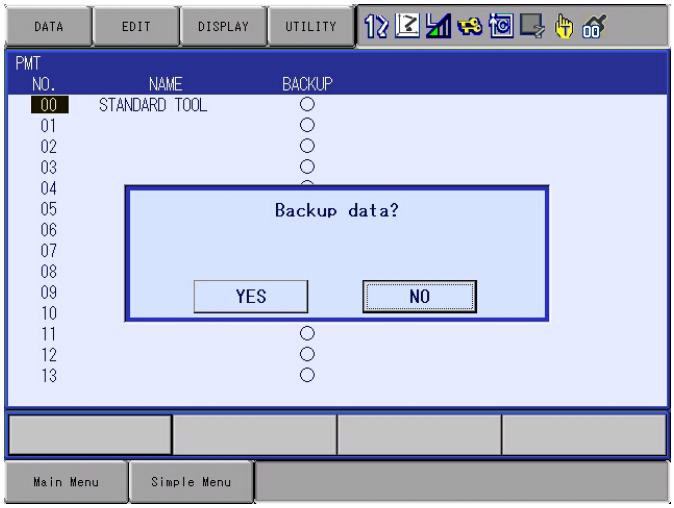
The following outline describes how to convert the data by using the programming pendant.

Refer to *chapter 2.3 "PMT Data Conversion by Programming Pendant"* for more information.

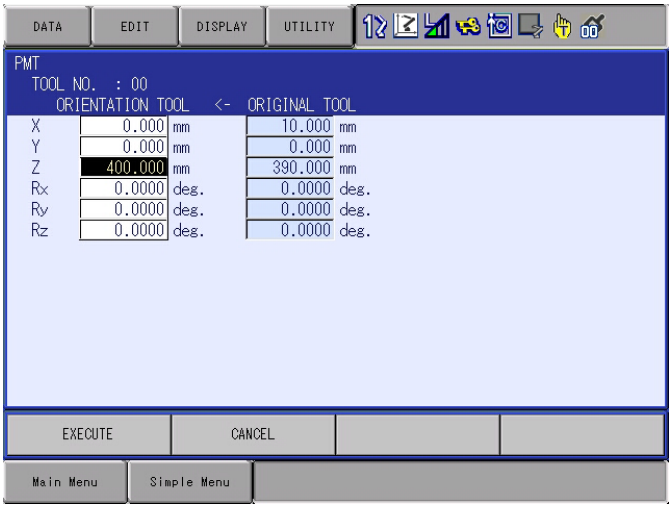
1. Select {PMT} under {UTILITY} in JOB CONTENT window.



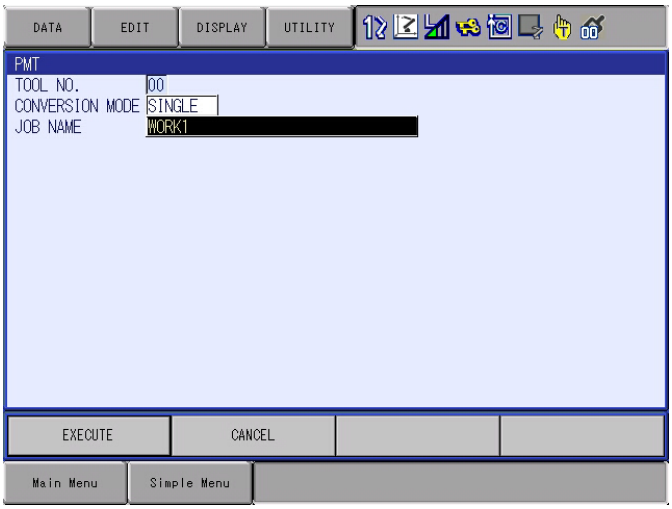
2. Save the tool constants used before the tool deformation as the original tool data.



3. Set the new tool constants to be used after the tool's deformation.



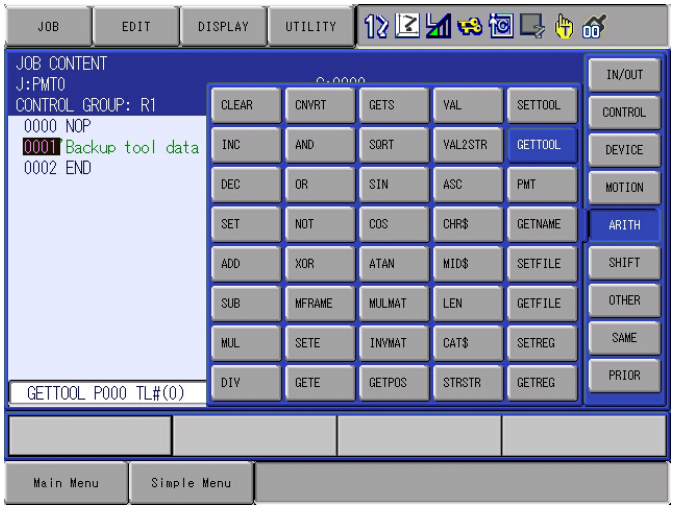
4. Convert the data of WORK1 using the PMT function.



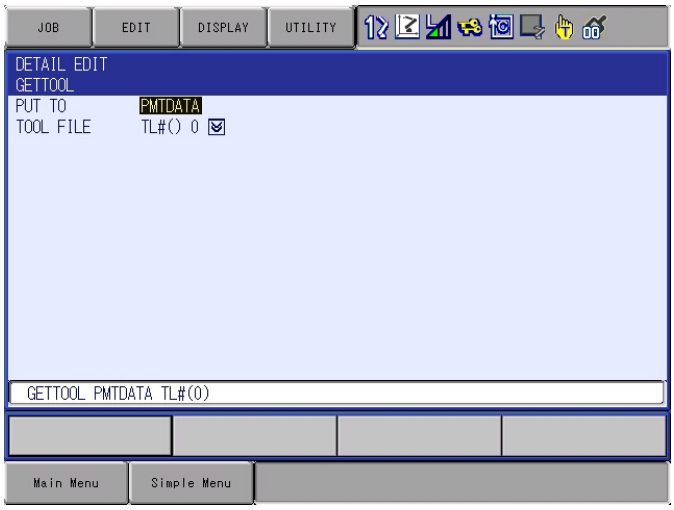
2.2 PMT Data Conversion by Instruction

2.2.1 Registering GETTOOL Instruction

1. Press [INFORM LIST]
2. Select the instruction group “ARITH”
3. Select “GETTOOL”
   – The instruction appears, with the same additional items that were previously set, in the input buffer line.



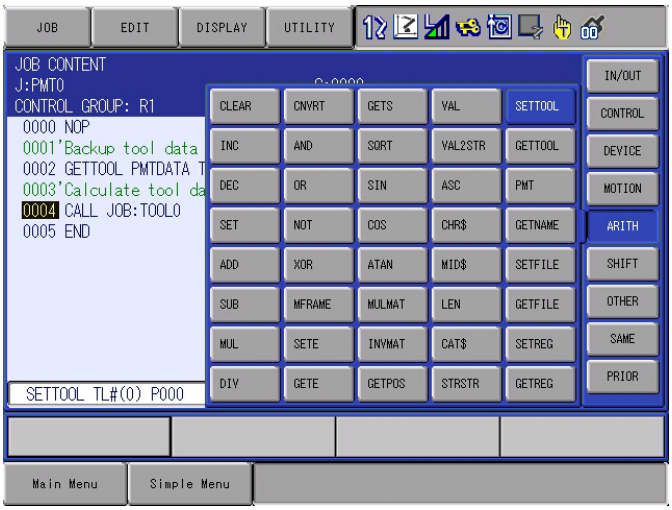
4. Select any additional items in the DETAIL EDIT window
   (1) Press [SELECT] two times, and the DETAIL EDIT window of the GETTOOL instruction appears.
   (2) Move the cursor to “PUT TO” and press [SELECT]. The selection dialog box appears.
   (3) Select “PMT DATA,” and the following window appears.



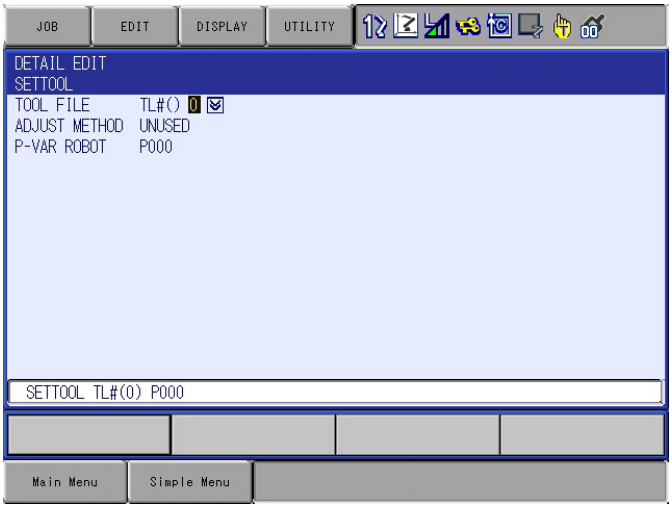
5. Press [ENTER]
   – The GETTOOL instruction with the additional items is added to the program for the job.
   – To cancel these settings, press [CANCEL], and the JOB CONTENT window reappears.

2.2.2 Registering SETTOOL Instruction

1. Press [INFORM LIST]
2. Select the instruction group “ARITH”
3. Select “SETTOOL”
   – The instruction appears, with the same additional items that were previously set, in the input buffer line.



4. Select any additional items in the DETAIL EDIT window
   (1) Press [SELECT] two times, and the DETAIL EDIT window of the SETTOOL instruction appears.
   (2) Set the TOOL FILE and the P-VAR ROBOT.

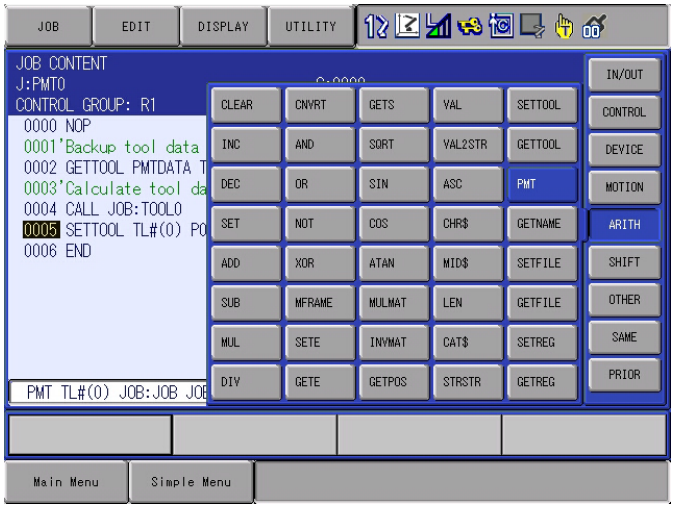


5. Press [ENTER]
   – The SETTOOL instruction is registered with the additional items.
   – To cancel these settings, press [CANCEL], and the JOB CONTENT window reappears.

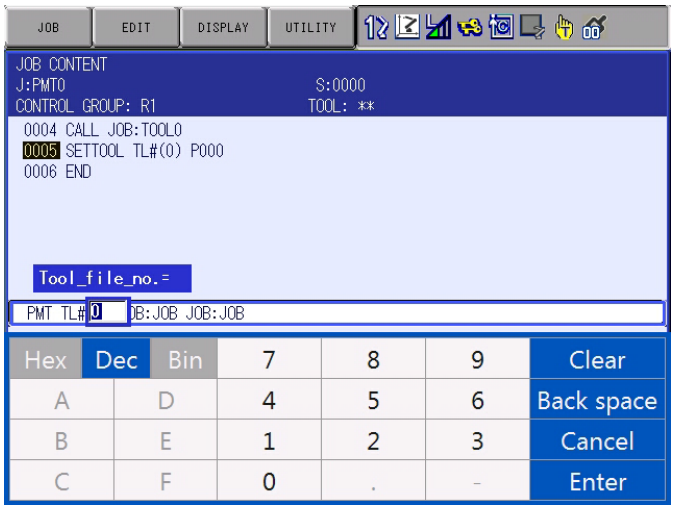
2.2.3 Registering PMT Instruction

NOTE Add PMT instructions for the number of jobs to be converted.

1. Press [INFORM LIST]
2. Select the instruction group “ARITH”
3. Select “PMT” instruction
   – The instruction appears, with the same additional items that were previously set, in the input buffer line.



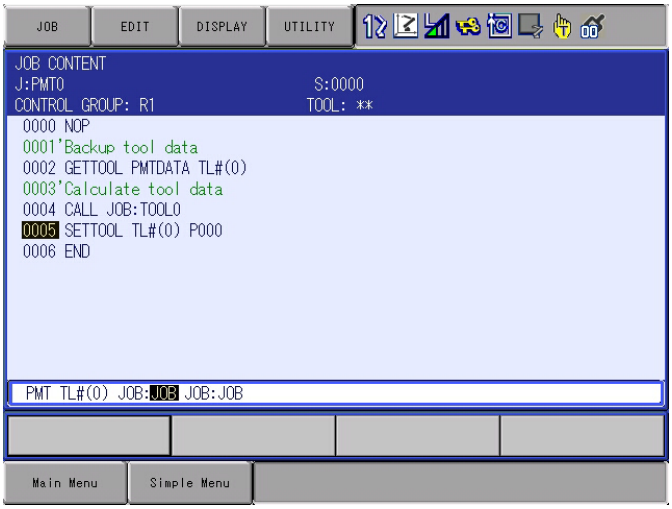
4. Select any additional items
   (1) Move the cursor to the tool file number and push [Select]. Then, it becomes possible to input the number by "Numeric Value."



(2) Move the cursor to the name of the job of which data is to be converted, and press [SELECT]. A list of jobs appears.

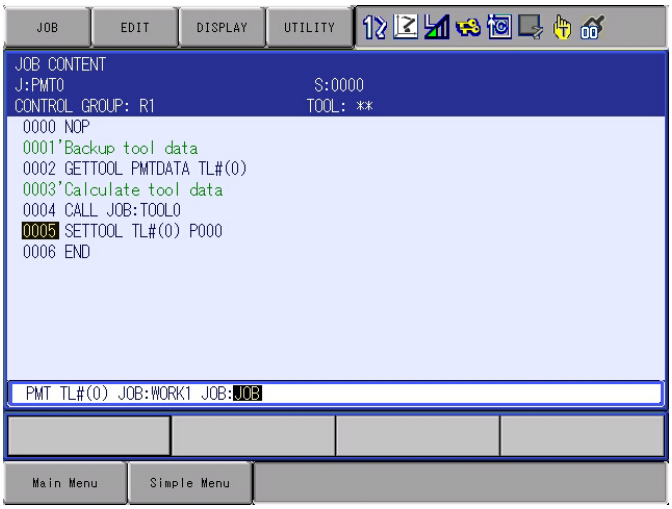
(3) Select the job subject to conversion for PMT function.

– Push "Page". Then, the screen changes to a text input screen. It is also possible to input a job name in the text input screen.



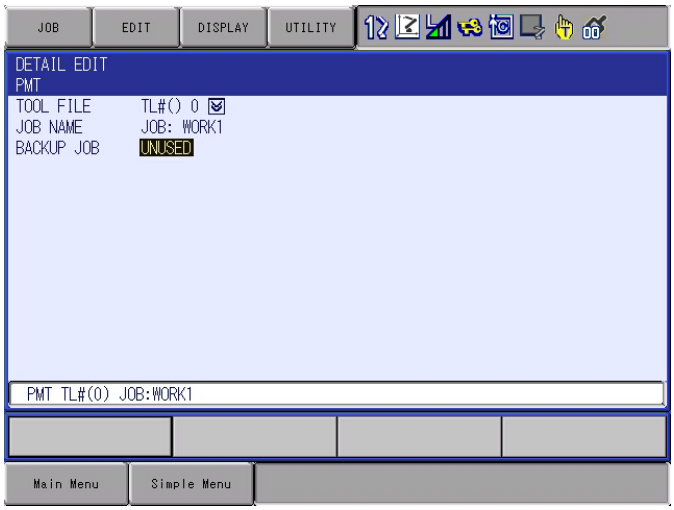
(4) Move the cursor to the name of the job to save the data, and press [SELECT]. A list of jobs appears.

– Push "Page". Then, the screen changes to a text input screen. It is also possible to input a job name in the text input screen.



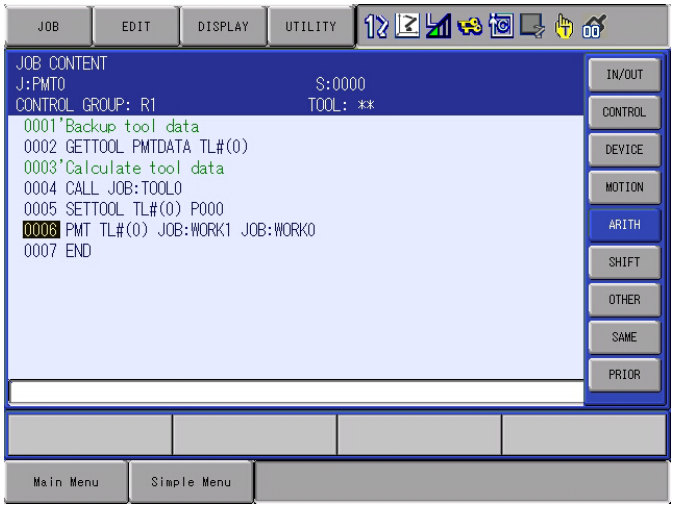
(5) When the name of the job to save the data does not have to be specified, move the cursor to “PMT,” and press [SELECT]. The DETAIL EDIT window appears.

– For the “BACKUP JOB,” select “UNUSED”.



5. Press [ENTER]

– The PMT instruction with the additional items is added to the program.



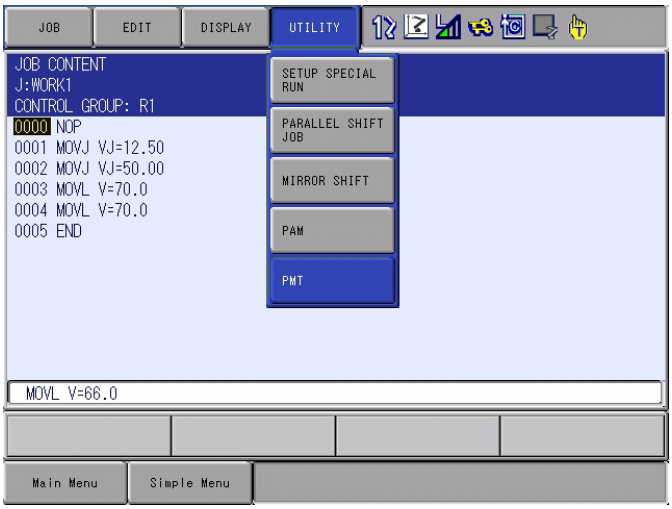
2.3 PMT Data Conversion by Programming Pendant

NOTE

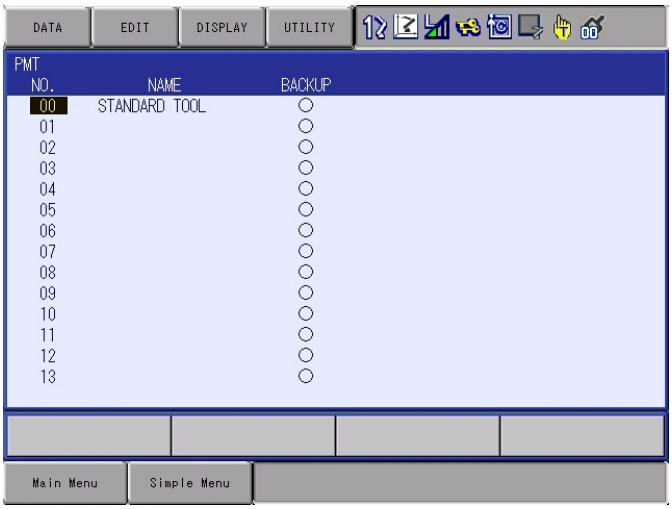
When the PMT function is used, the tool dimensions and the job taught position data are rewritten.

In order to secure the data, prepare the backup files beforehand.

1. Select {UTILITY} from the main menu in the job contents window in teach mode
   – The JOB CONTENT window appears.



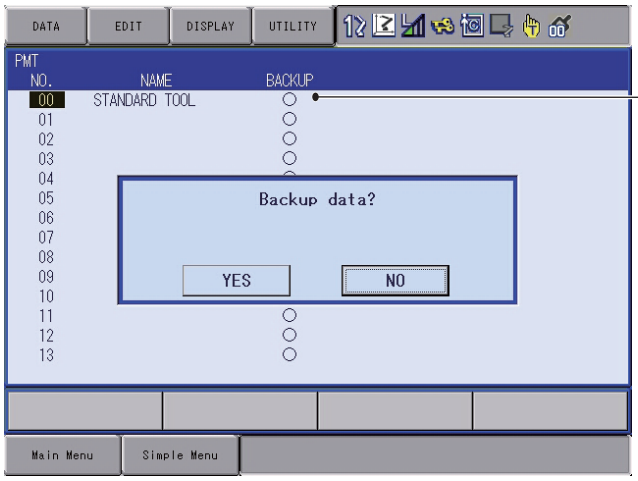
2. Select {PMT}
   – The PMT window appears.



3. Select No. of the tool for conversion

– The confirmation dialog to save the tool dimensions before deformation appears.

(1) Select "NO" if the tool dimensions after deformation are already registered.
Select "YES" if the tool dimensions after deformation are not registered yet.



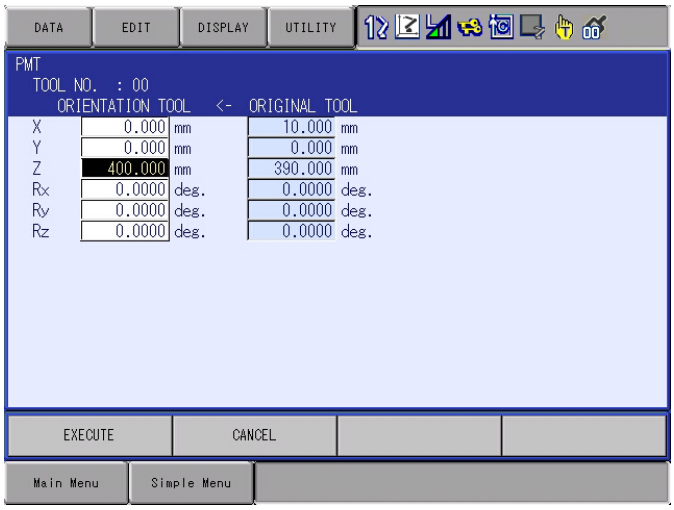
The indication of "●" would appear on the display if the registered tool demensions are the ones before deformation.

4. Register and confirm tool dimensions

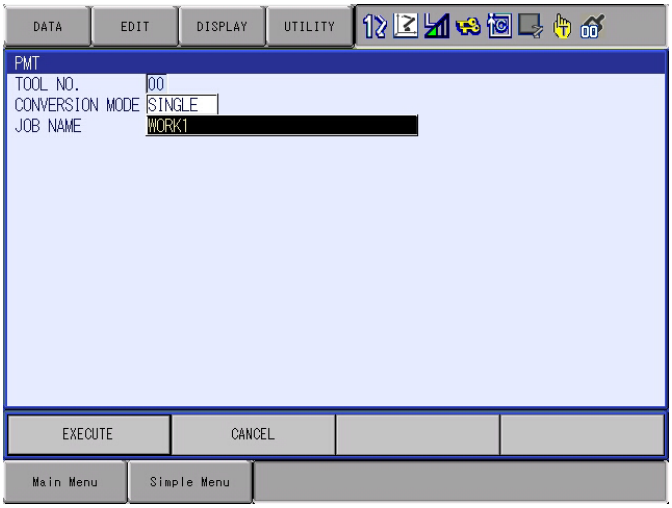
(1) Set the tool dimensions by inputting with the Numeric Keys or by selecting {TOOL CALIBRATION} of {UTILITY}.

– For details of methods to set tool dimensions, refer to "Chap. 8.3 Tool Data Setteing" in "YRC1000micro INSTRUCTIONS (RE-CTO-A222)".

(2) Confirm the tool dimensions before and after the PMT conversion (ORIGINAL TOOL and ORIENTATION TOOL), then select “EXECUTE”.



5. Select “CONVERSION MODE”.
   – The following three methods are available for selecting a job.



   • Method 1: SINGLE
     When only the specified job is to be converted, select “SINGLE” from “CONVERSION MODE”.

   • Method 2: RELATED
     When the jobs related to the specified job are to be converted together, select “RELATED” from “CONVERSION MODE”.

   • Method 3: ALL
     When all the jobs registered in YRC1000micro are to be converted,
     select “ALL” from “CONVERSION MODE”.

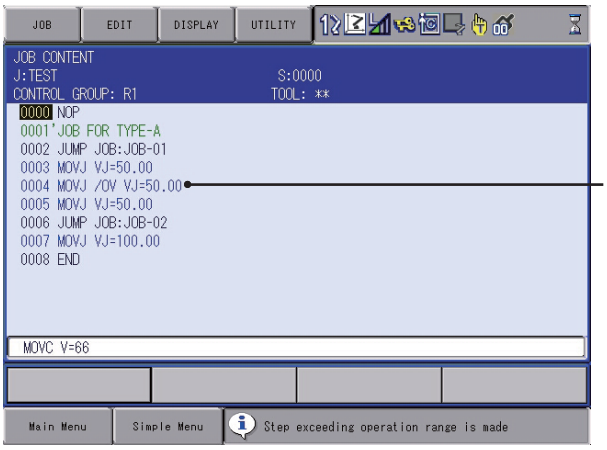
6. Select the job for conversion
   – Select “JOB NAME” to display the Job List, then select the job to be converted.

7. Select “EXECUTE”
   – Job conversion is performed based on the settings of conversion mode.
   – The window returns to the JOB CONTENT window when the conversion is completed.

NOTE

For a job after conversion, be sure to confirm the path after conversion by performing FWD/BWD operations.

If the steps beyond the manipulator's range of motion are created, "/0V" will be indicated in the corresponding step as shown in the following window. The "/0V" will disappear by correcting the positions.



In the step which is out of the manipulator's range of motion, "/OV" is indicated for a move instruction.

However, if any step beyond the scope of operation is created at the time of conversion, it is also possible to indicate an error and not to allow conversion by setting Parameter S2C390. (Refer to *chapter 5 "Parameter"* .)

In this case, "Error 2790: Step exceeding operation range (J: Job name, L: Line number)" is indicated.

3 Tool Data Backup History

The modification history of the tool constants can be viewed in the TOOL BACKUP window.

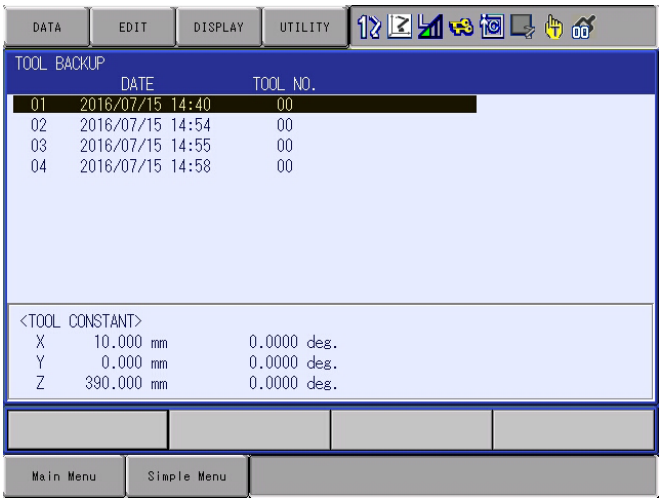
The history of the tool data backup is updated when a GETTOOL instruction with “PMTDATA” added is carried out.

When a GETTOOL instruction is carried out and the data of the original tool and the orientation tool are the same, only the execution date and time in the backup history are updated.

The backup history would be also updated when the original tool is saved in operating PMT function with the programming pendant.

3.1 Tool Backup Window

1. Select “ROBOT” under the main menu
2. Select {TOOL BACKUP}
   – The TOOL BACKUP window appears.



4 Instruction List for PMT Function

GETTOOL	Function	Gets the data from a tool file.	
	Additional items	Tool data storage directory P, PMTDATA[1)]	—
		Tool file TL# (<Tool file number>)	0 to 63
	Example	GETTOOL PMTDATA TL# (0) GETTOOL P000 TL# (0)	
SETTOOL	Function	Rewrites the data in the tool file.	
	Additional items	Tool file TL# (<Tool file number>)	0 to 63
		Adjustment method +, -	—
		Position variable robot P	—
	Example	SETTOOL TL# (0) P000 SETTOOL TL# (0) +P000	
PMT	Function	Converts the data.	
	Additional items	Tool file TL# (<Tool file number>)	0 to 63
		Name of the job of which data is subjected to conversion JOB:	
		Name of the job where the data is to be saved[2)] JOB:	
	Example	PMT TL# (0) JOB:WORK1 PMT TL# (0) JOB:WORK1 JOB:WORK0	

1. When “PMTDATA” is selected for the storage destination of the tool data, the specified data from the tool file is saved as the data of the backup tool.
2. When the name of the job specified as the job to save the data does not exist, the job used before converting the data (the job of which data is to be converted) is saved with the specified job name (JOB COPY).
   When the name of the job specified as the job to save the data already exists, the specified job is rewritten by the job used before converting the data.

5 Parameter

Parameter No.	Meaning	Units	Initial Value
S3C1192	The allowable difference (values of X, Y, and Z) between the present tool constant and the tool data automatic setting in the operation of a SETTOOL instruction.	0.001mm	20000
S2C390	0 : Conventional mode (conversion by /OV) 1 : Conversion after confirmation of limit checks at once (no conversion by /OV)	-	0

6 Alarm List

Alarm Number	Alarm Name	Sub Code	Meaning	Cause	Remedy
4605	SETTOOL ERROR	1	The difference between the current tool constant and a new set value exceeded the allowable range (parameter set value).	Setting error	(1)Check the following settings. ·Correct the job so that the setting value of tag is allowable value. ·Set the allowance amount of the tool data automatic setting function maximum deviation (S3C1192) to large value.
4126	CANNOT EXECUTE AUTO PMT	1	System error	Software operation error occurred	(1)Reset the alarm, and then try again. (2)If the alarm occurs again, save the CMOS.BIN in maintenance mode, and then contact your YASKAWA representative about occurrence status (operating procedure).
		2	PBOX cannot be edited.	Setting error	(1)Check the following settings. ·I/O status of the edit prohibit signal The edit prohibit signal cannot input.
		3	The source job cannot be edited.	Setting error	(1)Check the following settings. ·The prohibit status of source job If the source job is protected from editing, it cannot be edited.
		4	The converted job cannot be edited.	Setting error	(1)Check the following settings. ·The prohibit status of converted job If the converted job is protected from editing, it cannot be edited.
		5	The memory area for job area is insufficient.	Software operation error occurred	(1)Reset the alarm. (2)If the error occurs again, delete unused jobs. (3)If the error occurs again after the previous measures were executed, initialize the job file in the maintenance mode, and then load the saved job file. In that case, delete the unused jobs. (4)If the error occurs again though the previous measures were executed, save the CMOS.BIN in maintenance mode, and then contact your YASKAWA representative about occurrence status (operating procedure).

Alarm Number	Alarm Name	Sub Code	Meaning	Cause	Remedy
		6	The source job is not exist.	Setting error	(1)Check the following settings. ·Presence of the specified source job The job which does not exist cannot be set to the source job.
		7	The memory area for position data of the job is insufficient.	Software operation error occurred	(1)Reset the alarm. (2)when the error occurs again, if there is an unnecessary teaching position, delete it. (3)If the error occurs again after the previous measures were executed, initialize the job file in the maintenance mode, and then load the saved job file. In that case, delete the unused jobs. (4)If the error occurs again though the previous measures were executed, save the CMOS.BIN in maintenance mode, and then contact your YASKAWA representative about occurrence status (operating procedure).
		8	The job under execution is specified as the conversion job.	Setting error	(1)Check the following settings. ·Execution status of the source job ·Execution status of the converted job The job under execution is specified for the source / converted job. Execute conversion operation after ending the job execution.

YRC1000micro OPTIONS
INSTRUCTIONS

FOR PMT FUNCTION

HEAD OFFICE
2-1 Kurosakishiroishi, Yahatanishi-ku, Kitakyushu 806-0004, Japan
Phone +81-93-645-7703 Fax +81-93-645-7802

YASKAWA America Inc. (Motoman Robotics Division)
100 Automation Way, Miamisburg, OH 45342, U.S.A.
Phone +1-937-847-6200 Fax +1-937-847-6277

YASKAWA Europe GmbH（Robotics Divsion）
Yaskawastrasse 1, 85391 Allershausen, Germany
Phone +49-8166-90-100 Fax +49-8166-90-103

YASKAWA Nordic AB
Verkstadsgatan 2, Box 504 ,SE-385 25 Torsas, Sweden
Phone +46-480-417-800 Fax +46-486-414-10

YASKAWA Electric (China) Co., Ltd.
22F, One Corporate Avenue, No.222, Hubin Road, Huangpu District, Shanghai 200021, China
Phone +86-21-5385-2200 Fax ＋86-21-5385-3299

YASKAWA SHOUGANG ROBOT Co. Ltd.
No7 Yongchang North Road, Beijing E&T Development Area, China 100176
Phone +86-10-6788-2858 Fax +86-10-6788-2878

YASKAWA India Private Ltd. (Robotics Division)
#426, Udyog Vihar, Phase- IV, Gurgaon, Haryana, India
Phone +91-124-475-8500 Fax +91-124-475-8542

YASKAWA Electric Korea Corporation
35F, Three IFC, 10 Gukjegeumyung-ro, Yeongdeungpo-gu, Seoul, Korea 07326
Phone +82-2-784-7844 Fax +82-2-784-8495

YASKAWA Electric Taiwan Corporation
12F, No.207, Sec. 3, Beishin Rd., Shindian District, New Taipei City 23143, Taiwan
Phone +886-2-8913-1333 Fax +886-2-8913-1513

YASKAWA Electric (Singapore) PTE Ltd.
151 Lorong Chuan, #04-02A, New Tech Park, Singapore 556741
Phone +65-6282-3003 Fax +65-6289-3003

YASKAWA Electric (Thailand) Co., Ltd.
59,1st-5th Floor, Flourish Building, Soi Ratchadapisek 18,Ratchadapisek Road,
Huaykwang, Bangkok 10310, THAILAND
Phone +66-2-017-0099 Fax +66-2-017-0199

PT. YASKAWA Electric Indonesia
Secure Building-Gedung B Lantai Dasar & Lantai 1 Jl. Raya Protokol Halim Perdanakusuma,
Jakarta 13610, Indonesia
Phone +62-21-2982-6470 Fax +62-21-2982-6741

Specifications are subject to change without notice
for ongoing product modifications and improvements.

YASKAWA

YASKAWA ELECTRIC CORPORATION

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MANUAL NO.
HW1484458